Multi-Objective Optimisation and Multi-Criteria Decision-Making for Solar Green Hydrogen in Arid Environments: Cost, Reliability, Water and Land Trade-offs

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Chapter title: Multi-Objective Optimisation and Multi-Criteria Decision-Making for Solar Green Hydrogen in Arid Environments: Cost, Reliability, Water and Land Trade-offs

Target part: Part II – Multi-Objective Optimization and Multi-Criteria Decision-Making for Green Hydrogen Systems (also fits Part IV – Green Hydrogen in Arid, Desert, and Harsh Environments)

Abstract (5 sentences): Arid regions offer the best solar resource for green hydrogen, but dust, water scarcity and land use create trade-offs that single-objective cost studies overlook. This chapter sizes an off-grid PV–battery–PEM electrolyser–hydrogen plant with a three-objective NSGA-II that trades cost, supply shortfall and water intensity, treating the PV cleaning interval as a decision that balances soiling losses against cleaning water. TOPSIS then selects designs for four stakeholder weightings, and a statistical analysis over random seeds and 10 000 random weightings tests the robustness of both fronts and choices. Using open NREL data for a semi-arid and a humid site, the results show that reducing shortfall from 5 % to 1–2 % costs 1–1.7 USD/kg more, that reliability conflicts most strongly with water through module cleaning, and that a compromise near 8.2 USD/kg and 16.5 L/kg is achievable at both sites. The chapter closes with practical guidance for hydrogen deployment in dusty, water-scarce environments.

Abstract

Arid regions offer the best solar resource for green hydrogen, but dust, water scarcity and land use create trade-offs that single-objective cost studies overlook. This chapter formulates the sizing of an off-grid PV–battery–PEM electrolyser–hydrogen plant as a

three-objective problem — levelised cost of hydrogen (LCOH), supply shortfall and water intensity — with the PV cleaning interval as an explicit decision balancing soiling losses against cleaning water. A from-scratch NSGA-II, evaluated over ten weather years per site from open NREL data, finds Pareto fronts for a semi-arid and a humid site, and TOPSIS selects designs for four stakeholder weightings. Reducing shortfall from about 5 % to 1–2 % costs 1–1.7 USD/kg more, and the strongest conflict is between reliability and water, driven by module cleaning. The cost-led compromise reaches about 8.2 USD/kg with 4–5 % shortfall and 16.5 L/kg of water at both sites. Statistical analysis over seeds and 10 000 random weightings shows that NSGA-II outperforms random search, that the recommended design is robust at the humid site but sensitive to the water weight at the arid site, and that heavier dust shrinks the trade-off space while cleaning absorbs most of its cost.

Keywords: green hydrogen · multi-objective optimisation · NSGA-II · TOPSIS · multi-criteria decision-making · soiling · water–energy–hydrogen nexus · arid environments · hypervolume

1 Introduction

The sunniest places on Earth are also among the driest and dustiest. Arid regions offer the irradiance that makes solar hydrogen cheap, and many of the largest announced green hydrogen hubs are planned in deserts [1, 2]. The same environment imposes costs that single-objective cost studies tend to miss: dust settles on PV modules and steals output [7, 8]; cleaning it off consumes water; electrolysis itself consumes purified water [9, 10]; and every hectare of array needs land.

These effects create genuine trade-offs. Cleaning more often recovers PV output but uses scarce water. Building more PV raises reliability but also raises cost, land and cleaning water. No single design is best on cost, reliability and water at once; the planner must choose a compromise, ideally knowing the whole set of efficient alternatives — the Pareto front — and how the choice depends on their priorities.

Multi-objective evolutionary algorithms such as NSGA-II [3] find such fronts without weighting objectives in advance, and multi-criteria decision-making (MCDM) methods such as TOPSIS [4, 16] then select a design for a stated preference. Combining them for green hydrogen in harsh environments, with dust soiling and cleaning water as explicit decisions, and with a statistical check of how robust the front and the selection are, is the gap this chapter addresses.

The contributions are:

1. A **harsh-environment system model** that adds dust soiling, a PV cleaning interval as a decision variable, cleaning water and cost, and land use to an off-grid PV–battery–PEM–hydrogen plant.
2. A **three-objective NSGA-II** (written from scratch) that trades levelised cost of hydrogen (LCOH), supply shortfall and water intensity over ten weather years per site.
3. **TOPSIS decision-making** under four stakeholder weightings, plus a Monte Carlo weight-sensitivity analysis showing how often each design is chosen.
4. A **statistical evaluation**: hypervolume across random seeds, a comparison with random search, objective correlations on the front, and dust-intensity scenarios for the arid site, contrasted with a humid site.

Section 2 reviews background; Section 3 presents the model; Section 4 the optimisation and decision methods; Section 5 the results; Section 6 the statistical analysis; Section 7 discusses deployment in arid environments; Section 8 concludes.

2 Background

Multi-objective optimisation. A design is Pareto-optimal if no objective can be improved without worsening another. NSGA-II [3] approximates the set of such designs with a population that is sorted into non-dominated fronts and spread along them by a crowding-distance measure; variation uses simulated binary crossover [6] and polynomial mutation. The quality of an approximate front is commonly measured by its hypervolume — the volume of objective space it dominates up to a reference point [5] — which rewards both closeness to the true front and spread along it.

Multi-criteria decision-making. TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) [4] ranks alternatives by their relative closeness to an ideal point and distance from an anti-ideal point after vector normalisation and weighting. It is among the most widely applied MCDM methods in energy planning [16]. Its known weakness is dependence on the chosen weights, which motivates sensitivity analysis.

Soiling in arid regions. Dust deposition reduces PV output by fractions of a percent per day in deserts, and much more during dust storms; losses of 15–25 % accumulate within weeks without cleaning [7, 8]. Cleaning restores output but costs labour, equipment and — for wet methods — water. Ilse et al. [8] showed that the optimal cleaning frequency is a site-specific economic trade-off.

The water–energy–hydrogen nexus. Stoichiometry requires 9 L of water per kg of hydrogen; purification losses and cooling raise practical consumption towards 15–30 L/kg [9, 10]. In arid regions this water often comes from desalination, which adds energy and cost, though studies conclude that desalination cost is a small fraction of LCOH [11]. Land use of utility PV is roughly 2–3 hectares per MW of capacity [21], driven by row spacing.

Taken together, these strands suggest three questions this chapter answers quantitatively: how steeply do cost, reliability and water trade off; does cleaning pay at dusty sites; and how stable is the recommended design when preferences change?

3 System Model with Harsh-Environment Terms

3.1 Plant, sites and weather

The plant is an off-grid PV array, Li-ion battery, PEM electrolyser (minimum load 10 %, SEC = 48 + 6 l kWh/kg plus 2.5 kWh/kg compression) and compressed hydrogen tank serving a constant demand of 100 kg/day; any shortfall is bought at 10 USD/kg. A rule-based energy management system runs hourly: PV feeds the electrolyser first, surplus charges the battery, the battery tops up the electrolyser, and the rest is curtailed. PV output is modelled for a latitude-tilted, south-facing array with the Erbs decomposition [17] and pvlib [14].

Two sites with open NREL data are compared: **site A**, a semi-arid high desert in Colorado (NSRDB PSM3 2017 [12]; 1 749 kWh/m²·yr) and **site B**, a humid subtropical site in North Carolina (TMY3 723170 [13]; 1 566 kWh/m²·yr). Each design is evaluated on ten weather years per site: the observed year and nine seasonal moving-block-bootstrap years built from 5-day blocks [15]. All objectives are averages over the ten years.

3.2 Dust soiling and cleaning

Soiling loss grows linearly between cleanings at a rate r (% of PV output per day) and is reset by each wet cleaning every τ days, with a cap of 25 % for very long intervals. The mean loss over a cleaning cycle is:

```
\bar L(\tau) = \frac{1}{\tau}\sum_{j=0}^{\tau-1} \min(r\,j,\; 25\,\%)
\;\approx\; \frac{r\,(\tau-1)}{2}
```

The soiled array is simulated as an array of capacity C_pv (1 − L̄). The base soiling rate is 0.25 %/day at the arid site and 0.05 %/day at the humid site, where frequent rain cleans the modules naturally; scenarios of 0.10 and 0.50 %/day bracket arid sites from mild to dusty [7, 8]. These rates are representative values from the literature, not measurements at the two sites.

3.3 Water, land and cost

Module area is 5 m² per kWp (about 20 % efficient modules) and each wet cleaning uses 0.5 L/m² and costs 0.015 USD/m² in labour and equipment. Electrolysis consumes 15 L of water per kg of hydrogen including purification losses. Water costs 3.0 USD/m³ at the arid

site (treatment and haulage, or desalination) and 1.5 USD/m³ at the humid site. Land is computed from module area at a ground-coverage ratio of 0.4, i.e. 1.25 ha per MWp of array (fenced area including roads is typically about twice this [21]).

Component	Capital cost	Fixed O&M	Life (yr)	Decision range
PV array	650 USD/kWp	15 USD/kWp·yr	25	100–4 000 kWp
PEM electrolyser	1 100 USD/kW	3 %/yr	20	50–2 000 kW
Li-ion battery	280 USD/kWh	2 %/yr	12	0–2 000 kWh
H_2 tank	500 USD/kg	1 %/yr	25	10–3 000 kg
PV cleaning interval	0.015 USD/m² per cleaning	–	–	1–90 days

Capital is annualised at 7 %. All cost figures are indicative 2025-level assumptions.

4 Multi-Objective Optimisation and Decision-Making

4.1 Problem

The decision vector is x = (C_pv, C_el, C_b, C_H, τ): PV capacity, electrolyser rating, battery energy, tank size and PV cleaning interval. Three objectives are minimised, each averaged over the ten weather years y:

```
f_1(x) = \frac{1}{10}\sum_{y} \frac{\mathrm{AC}(x) + c_{cl}\,\frac{365}
{\tau}\,A(x) + c_w\bigl(W^{feed}_y + W^{cl}\bigr) + c_{bk}\,U_y(x)}
{365\,D}\quad[\mathrm{USD/kg}]
```

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f_2(x) = \frac{1}{10}\sum_{y} \frac{U_y(x)}{365\,D}\times 100\;[\%], \qquad
f_3(x) = \frac{1}{10}\sum_{y} \frac{W^{feed}_y + W^{cl}}{M_y(x)}\;
[\mathrm{L/kg}]
```

where AC is the annualised capital and fixed O&M cost, A the module area, W^feed the electrolysis water, W^cl = (365/τ) · 0.5 L/m² · A the cleaning water, U_y the unmet hydrogen and M_y the hydrogen produced on site. f_3 is defined per kilogram produced on site so that buying backup hydrogen cannot make a design look water-efficient.

4.2 NSGA-II settings

NSGA-II [3] was implemented from scratch in NumPy: fast non-dominated sorting, crowding distance, binary tournament on (rank, crowding), simulated binary crossover (probability 0.9, distribution index 15) [6] and polynomial mutation (probability 1/5 per gene, index 20). Population 60 and 60 generations give 3 660 evaluations per run; each evaluation simulates 10 × 8 760 plant-hours, vectorised over the population. The implementation was verified on the ZDT1 benchmark, reaching a hypervolume of 0.714 against a theoretical 0.725.

Each base case was run with five random seeds; the dust scenarios with two. Hypervolume is estimated by Monte Carlo (200 000 points) in the box from (5 USD/kg, 0 %, 14 L/kg) to the reference point (12 USD/kg, 30 %, 40 L/kg), reported as a fraction of the box. A random-search baseline draws the same number of designs (3 660) uniformly and keeps its non-dominated set.

4.3 TOPSIS

TOPSIS [4] ranks the designs on the final front. The decision matrix of the three objectives is vector-normalised, multiplied by weights w, and each design's closeness $C = d^-/(d^+ + d^-)$ to the ideal point is computed, where d^+ and d^- are the Euclidean distances to the ideal and anti-ideal points. Four stakeholder weightings are examined:

Weighting	Cost (f_1)	Shortfall (f_2)	Water (f_3)	Typical stakeholder
Cost-led	0.5	0.3	0.2	Investor
Equal	1/3	1/3	1/3	Neutral planner
Reliability-led	0.3	0.5	0.2	Industrial off-taker
Water-led	0.3	0.2	0.5	Water authority in an arid region

To test robustness, 10 000 weight vectors are drawn uniformly from the simplex (Dirichlet(1, 1, 1)); for each, the TOPSIS winner is recorded. The share of draws a design wins, and the Spearman rank correlation between rankings under random weights and under the cost-led weights, measure how sensitive the recommendation is to preferences.

5 Results

The Pareto fronts show a steep price for the last few percent of reliability, a strong conflict between reliability and water, and a balanced compromise at about 8.2 USD/kg, 4–5 % shortfall and 16.5 L/kg at both sites.

5.1 Pareto fronts

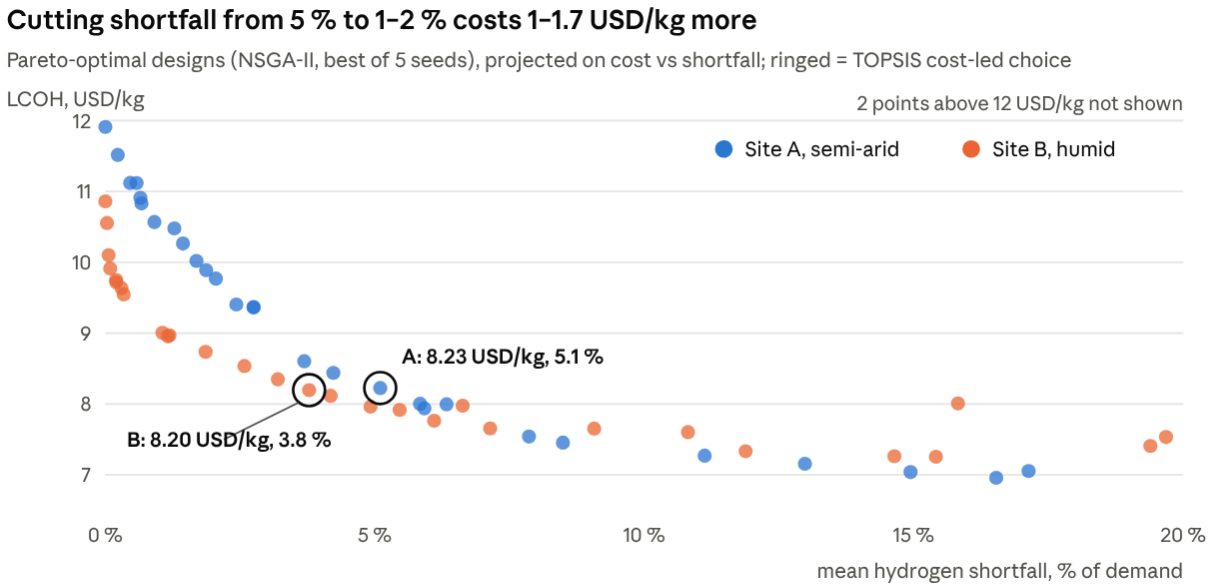


Fig. 1 · results/ch3_fronts.csv · base soiling, designs with shortfall ⩽ 20 %

Each front holds 60 non-dominated designs. Below about 5 % shortfall the fronts bend upwards: at site A, going from 5.1 % to 1.9 % shortfall raises LCOH from 8.23 to 9.89 USD/kg (0.51 USD/kg per point), and full reliability costs 11.9 USD/kg. The humid site's front is flatter (0.27 USD/kg per point from 4.9 % to 1.1 %), because its weaker but more even winter makes reliability cheaper to buy than the arid site's deep January trough. The cheapest designs on either front (6.96 and 7.26 USD/kg) accept 15–17 % shortfall.

Front property (base soiling)	Site A, semi-arid	Site B, humid
Lowest LCOH on front (USD/kg)	6.96	7.26
Lowest LCOH with shortfall ⩽ 5 % (USD/kg)	8.44	7.97
Water intensity range (L/kg produced)	15.4 – 28.9	15.4 – 17.7
Cleaning interval, 10 % / median / 90 % quantile (days)	8 / 90 / 90	28 / 90 / 90
Mean soiling loss, median design	11.1 %	2.2 %

Water intensity varies far more at the arid site: frequent cleaning (every 4–9 days) lifts total intensity to 22–29 L/kg, i.e. cleaning adds 7–14 L to the 15 L of electrolysis water per kilogram, whereas the humid site never exceeds 17.7 L/kg. Cleaning is used selectively: about half of the arid designs do not clean at all within the 90-day limit and accept an 11 % soiling loss, while the low-shortfall designs clean every 1–4 weeks.

5.2 Designs chosen by TOPSIS

Site	Weighting	PV (kWp)	EL (kW)	Battery (kWh)	Tank (kg)	Clean every (days)	LCOH (USD/kg)	Shortfall	Water (L/kg)	Soiling loss	Land (ha)
A	Cost-led, equal, water-led	1 808	966	276	144	31	8.23	5.1 %	16.5	3.8 %	2.26
A	Reliability-led	2 256	812	187	375	26	8.60	3.7 %	17.2	3.1 %	2.82
B	Cost-led, equal	1 964	936	195	168	36	8.20	3.8 %	16.4	0.9 %	2.46
B	Reliability-led	2 252	890	52	559	54	8.74	1.9 %	16.1	1.3 %	2.82
B	Water-led	1 967	786	0	111	90	7.66	7.1 %	15.6	2.2 %	2.46

At the arid site three of the four weightings pick the same design: about 1.8 MWp of PV and 1 MW of electrolyser, cleaned monthly, which keeps soiling under 4 % for only 1.5 L/kg of extra water. A reliability-minded off-taker pays 0.37 USD/kg more for 1.4 points less shortfall, mainly through a 2.6-fold larger tank. At the humid site the choice spreads more: a water authority would skip the battery and cleaning altogether, accepting 7 % shortfall at a lower LCOH of 7.66 USD/kg. Land for the compromise designs is 2.3–2.8 ha of array (about twice that fenced).

6 Statistical Analysis

NSGA-II finds better and more consistent fronts than random search, but the margin is modest; the TOPSIS recommendation is stable at the humid site and fragile at the arid site; and stronger dust shrinks the attainable trade-off space monotonically.

6.1 Front quality: NSGA-II versus random search

Site	NSGA-II hypervolume, mean ± SD (5 seeds)	Random search, mean ± SD (5 seeds)	Difference	Wilcoxon signed-rank, one-sided p (paired by seed)	Mann–Whitney U, one-sided p
A, semi-arid	0.566 ± 0.006	0.557 ± 0.009	+0.008	0.031	0.048
B, humid	0.583 ± 0.006	0.555 ± 0.019	+0.027	0.063	0.008

Hypervolume is the fraction of the objective box dominated by each front. NSGA-II beats random search at both sites, significantly at the 5 % level by at least one test at each site; with five seeds the smallest attainable one-sided Wilcoxon p-value is 1/32 = 0.031, so site

A reaches the floor and site B misses it because random search won one seed. Two further points matter in practice. First, NSGA-II's seed-to-seed spread is one-third of random search's at site B (SD 0.006 vs 0.019), so a single NSGA-II run is more trustworthy. Second, random search is a strong baseline here because each evaluation is cheap and the problem has only five decisions; the gap would widen with more decisions or tighter evaluation budgets.

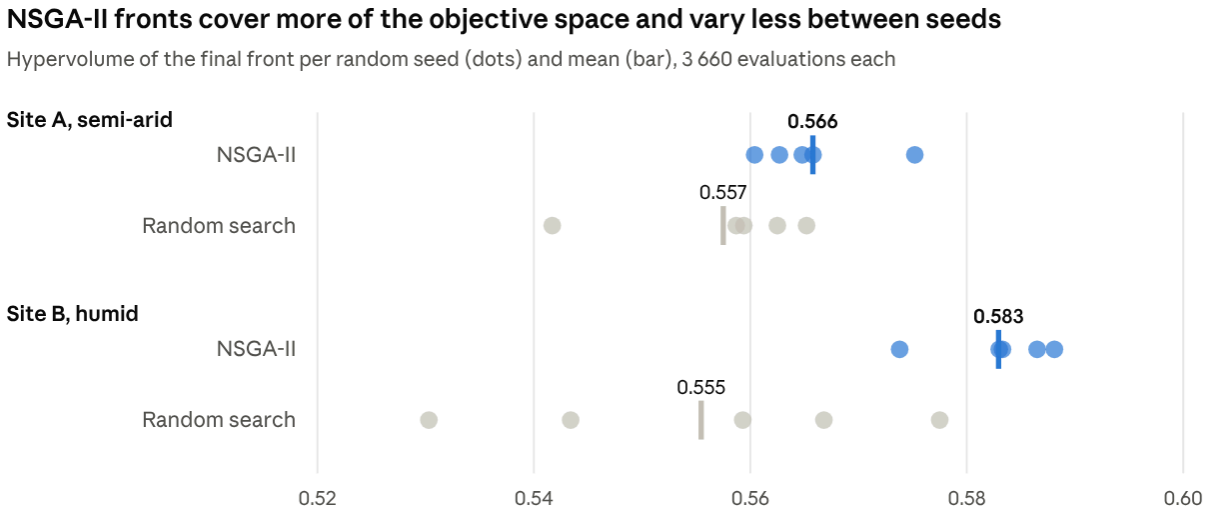


Fig. 2 · results/ch3_hypervolume.csv · 5 seeds per method and site

6.2 How the objectives conflict

Spearman rank correlations between objectives across the 60 designs on each front quantify the trade-offs:

Objective pair	Site A	Site B	Reading
Shortfall vs water	−0.89	−0.90	strong conflict: reliable designs use more water
LCOH vs water	−0.69	−0.73	strong conflict: water-lean designs cost more
LCOH vs shortfall	+0.50	+0.58	partly aligned: very unreliable designs also cost more, because backup hydrogen is bought at 10 USD/kg

The dominant conflict is reliability versus water, not cost versus reliability. Electrolysis water per kilogram is fixed at 15 L, so the conflict comes entirely from cleaning: low-shortfall designs clean more often to recover PV output (every 1–4 weeks rather than not at all), and their larger arrays have more module area to wash per kilogram produced. The positive LCOH–shortfall correlation is a reminder that with a backup price near the LCOH, the cheapest designs are not the least reliable ones.

6.3 Robustness of the TOPSIS choice

Across 10 000 random weight vectors drawn uniformly from the simplex:

Site	Distinct designs ever chosen (of 60)	Share won by the most frequent design	Share won by top 3 designs	Spearman ρ of rankings vs cost-led weights, median (5 % quantile)
A, semi-arid	23	20 %	53 %	0.87 (0.09)
B, humid	26	25 %	52 %	0.90 (0.76)

The top three designs win about half of all weightings at both sites, so a short list is robust even when a single winner is not. The full ranking is stable at the humid site (ρ ≥ 0.76 in 95 % of weightings) but can reverse almost completely at the arid site (5 % quantile ρ = 0.09). The reason is the wider water range at the arid site: when a weighting emphasises water, designs that skip cleaning jump up the ranking. Decision-makers at dusty sites should therefore agree on the water weight explicitly before choosing.

6.4 Dust-intensity scenarios at the arid site

Soiling rate	Seeds	Hypervolume, mean	Median cleaning interval on front (days)	Median soiling loss	Maximum water intensity (L/kg)
Low, 0.10 %/day	2	0.576	90	4.5 %	23.9
Base, 0.25 %/day	5	0.566	90	11.1 %	28.9
High, 0.50 %/day	2	0.547	35	8.5 %	28.5

As dust increases, the attainable trade-off space shrinks monotonically (hypervolume 0.576 → 0.566 → 0.547) and the optimiser switches from tolerating soiling to cleaning: the median front design cleans every 35 days under heavy dust instead of not cleaning at all. The minimum LCOH at ≤ 5 % shortfall varied non-monotonically (8.31, 8.44 and 8.35 USD/kg), which is within the seed-to-seed variability of the fronts; so within this range dust changes how the plant is run more than what its best compromise costs, because cleaning offsets most of the loss at a cost of extra water.

6.5 Which decisions drive which objective

PV capacity is the single lever that moves designs along the front: it explains almost all of the variation in shortfall and in water intensity, while the electrolyser and battery mainly move cost.

Spearman correlations between each decision variable and each objective were computed across the 60 designs of each base front, together with the median and interquartile range (IQR) of each decision.

Decision	Site A: ρ with LCOH / shortfall / water	Site B: ρ with LCOH / shortfall / water	Site A: median [IQR]	Site B: median [IQR]
PV capacity (kWp)	−0.54 / **−0.97** / **+0.92**	−0.60 / **−0.99** / **+0.91**	1 110 [653, 2 268]	1 171 [684, 2 017]
Electrolyser (kW)	**+0.77** / +0.11 / −0.35	**+0.77** / +0.14 / −0.32	927 [814, 1 713]	940 [863, 1 609]
Battery (kWh)	**+0.80** / +0.71 / −0.78	**+0.73** / +0.76 / −0.82	286 [145, 1 102]	286 [149, 698]
Tank (kg)	+0.54 / −0.28 / +0.03	+0.53 / −0.10 / −0.02	449 [166, 1 816]	460 [178, 582]
Cleaning interval (days)	+0.64 / +0.79 / **−0.89**	+0.56 / +0.78 / **−0.82**	90 [15, 90]	90 [47, 90]

Three patterns are consistent across both climates. (i) PV capacity correlates almost perfectly with shortfall (ρ = −0.97 and −0.99): reliability is bought with array size. (ii) Water intensity is driven by PV size and cleaning frequency together (ρ = +0.92 and −0.89 at the arid site), confirming that the reliability–water conflict runs through module area and washing. (iii) Larger batteries appear in cheap, unreliable, water-lean designs (positive correlation with both LCOH and shortfall): along this front, battery capacity substitutes for PV rather than adding reliability — a battery shifts energy within a day but cannot create the energy a too-small array lacks in winter. Among the more reliable designs (shortfall ⩽ 10 %), the PV-to-electrolyser ratio is a stable 2.4–2.5 and the tank holds 115–140 hours (about five to six days) of demand — two practical rules of thumb for early-stage sizing at these sites.

7 Discussion: Deploying Green Hydrogen in Arid and Harsh Environments

Water is a reliability question more than a cost question. Across both fronts, water never moved LCOH by more than a few cents per kilogram at 1.5–3 USD/m³. What water *does* constrain is reliability: the designs that deliver hydrogen most reliably are the ones that wash their modules most often. Where water rights are capped — common in deserts — the planner should expect to buy reliability with more PV, more storage or a contracted backup rather than with cleaning water. Dry or robotic cleaning, which uses little or no water [7, 8], decouples the two and would shift the arid front towards the humid one.

A cleaning interval is a design decision. Treating cleaning as an optimisation variable rather than a fixed O&M assumption changed the answer: under base dust most efficient designs do not clean at all, the compromise designs clean monthly, and heavy dust pushes the typical interval to about five weeks. Fixed assumptions such as "clean every two weeks" would waste water at mild sites and lose output at dusty ones.

Present a short list, not one design. The TOPSIS winner changed with weights in about four of five random weightings, yet the top three designs covered half of them. Presenting stakeholders with three well-separated designs — cost-led, reliability-led and water-led — is more honest and more useful than one "optimal" plant. At dusty sites, agreeing the water weight first is essential because rankings can invert.

Land. The compromise designs need 2.3–2.8 ha of array, roughly 5 ha fenced, for 100 kg/day. Scaled linearly to a 100 t/day hub this is in the order of 50 km² — available in many deserts but a real constraint near cities, farmland or protected habitats.

Limitations. (i) Soiling rates are literature values, not site measurements, and soiling is averaged over each cycle rather than simulated with dust events and rain. (ii) Only one year of observed data per site was available; block-bootstrap years add variability but not inter-annual climate variation. (iii) High ambient temperature effects on electrolyser cooling and efficiency, and sand ingress into balance-of-plant, were not modelled. (iv) Costs are indicative. (v) Hypervolume comparisons use five seeds, which limits statistical power; more seeds would sharpen the NSGA-II versus random-search comparison.

Future directions. Coupling the optimiser to measured soiling and rainfall series, adding dry-cleaning technologies and desalination plant sizing as decisions, and replacing the simulator with a learned surrogate to allow many more objectives (land, embodied carbon, local employment) are natural next steps.

8 Conclusion

This chapter optimised an off-grid solar hydrogen plant for cost, reliability and water at a semi-arid and a humid site, with PV cleaning as an explicit decision. A from-scratch NSGA-II produced 60-design Pareto fronts that beat random search in hypervolume (0.566 vs 0.557 and 0.583 vs 0.555) with a third of its seed-to-seed spread at the humid site. The fronts show that cutting shortfall from about 5 % to 1–2 % costs 1–1.7 USD/kg more, and that the strongest conflict is between reliability and water (Spearman ρ = −0.89 to −0.90), driven by module cleaning. TOPSIS selected a compromise near 8.2 USD/kg, 4–5 % shortfall and 16.5 L/kg at both sites, with monthly cleaning at the arid site; the choice was stable at the humid site but sensitive to the water weight at the arid site. Heavier dust shrank the attainable trade-off space monotonically and made cleaning worthwhile, while leaving the best compromise cost within optimiser noise. For arid hydrogen hubs, the practical messages are to treat cleaning as a design variable, to settle the water weight with stakeholders up front, and to present a short list of designs rather than a single optimum.

Code and data availability. All code (NSGA-II, TOPSIS, plant simulator) and data are in the Book-Chapter-Green-Hydrogen folder of the authors' repository.

References

References are given from the authors' knowledge (no live database lookup was possible); check every entry and convert to Springer style before submission.

1. IRENA (2020) Green hydrogen cost reduction: Scaling up electrolysers to meet the 1.5 °C climate goal. International Renewable Energy Agency, Abu Dhabi.
2. IEA (2019) The future of hydrogen: Seizing today's opportunities. International Energy Agency, Paris.
3. Deb K, Pratap A, Agarwal S, Meyarivan T (2002) A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Transactions on Evolutionary Computation 6(2):182–197.
4. Hwang CL, Yoon K (1981) Multiple attribute decision making: Methods and applications. Springer, Berlin.
5. Zitzler E, Thiele L (1999) Multiobjective evolutionary algorithms: A comparative case study and the strength Pareto approach. IEEE Transactions on Evolutionary Computation 3(4):257–271.
6. Deb K, Agrawal RB (1995) Simulated binary crossover for continuous search space. Complex Systems 9(2):115–148.
7. Sarver T, Al-Qaraghuli A, Kazmerski LL (2013) A comprehensive review of the impact of dust on the use of solar energy: History, investigations, results, literature, and mitigation approaches. Renewable and Sustainable Energy Reviews 22:698–733.

8. Ilse K, Micheli L, Figgis BW, Lange K, Dassler D, Hanifi H, Wolfertstetter F, Naumann V, Hagendorf C, Gottschalg R, Bagdahn J (2019) Techno-economic assessment of soiling losses and mitigation strategies for solar power generation. Joule 3(10):2303–2321.
9. Beswick RR, Oliveira AM, Yan Y (2021) Does the green hydrogen economy have a water problem? ACS Energy Letters 6(9):3167–3169.
10. Shi X, Liao X, Li Y (2020) Quantification of fresh water consumption and scarcity footprints of hydrogen from water electrolysis: A methodology framework. Renewable Energy 154:786–796.
11. Khan MA, Al-Attas T, Roy S, Rahman MM, Ghaffour N, Thangadurai V, Larter S, Hu J, Ajayan PM, Kibria MG (2021) Seawater electrolysis for hydrogen production: A solution looking for a problem? Energy & Environmental Science 14:4831–4839.
12. Sengupta M, Xie Y, Lopez A, Habte A, Maclaurin G, Shelby J (2018) The National Solar Radiation Data Base (NSRDB). Renewable and Sustainable Energy Reviews 89:51–60.
13. Wilcox S, Marion W (2008) Users manual for TMY3 data sets. Technical Report NREL/TP-581-43156, National Renewable Energy Laboratory, Golden, CO.
14. Holmgren WF, Hansen CW, Mikofski MA (2018) pvlib python: A python package for modeling solar energy systems. Journal of Open Source Software 3(29):884.
15. Künsch HR (1989) The jackknife and the bootstrap for general stationary observations. The Annals of Statistics 17(3):1217–1241.
16. Behzadian M, Otaghsara SK, Yazdani M, Ignatius J (2012) A state-of-the-art survey of TOPSIS applications. Expert Systems with Applications 39(17):13051–13069.
17. Erbs DG, Klein SA, Duffie JA (1982) Estimation of the diffuse radiation fraction for hourly, daily and monthly-average global radiation. Solar Energy 28(4):293–302.
18. Wilcoxon F (1945) Individual comparisons by ranking methods. Biometrics Bulletin 1(6):80–83.
19. Mann HB, Whitney DR (1947) On a test of whether one of two random variables is stochastically larger than the other. The Annals of Mathematical Statistics 18(1):50–60.
20. Goldberg DE (1989) Genetic algorithms in search, optimization and machine learning. Addison-Wesley, Reading, MA.
21. Ong S, Campbell C, Denholm P, Margolis R, Heath G (2013) Land-use requirements for solar power plants in the United States. Technical Report NREL/TP-6A20-56290, National Renewable Energy Laboratory, Golden, CO.